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ARC PATH FORMATION PART AND DIRECT CURRENT RELAY COMPRISING SAME

Abstract

The present invention discloses an arc path formation unit and a direct current relay including the same, which can effectively guide a generated arc to the outside, including a magnet holder unit disposed between the outside of an arc chamber and the inside of a frame and including a first holder and a second holder different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent at a predetermined angle and extended, the magnet unit is attached to both ends thereof, and a magnetic field formed in the magnet unit forms an electromagnetic force together with the electric current energizing through the direct current relay to guide the arc in a direction away from a fixed contact.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a National Stage of International Application No. PCT/KR2022/017910, filed on Nov. 14, 2022, which claims priority to and the benefit of Korean Patent Application No. 10-2021-0159319, filed on Nov. 18, 2021, the disclosures of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] The present disclosure relates to an arc path formation unit and a direct current relay including the same, and more particularly, to an arc path formation unit having a structure capable of effectively guiding a generated arc to the outside, and a direct current relay including the same.
BACKGROUND
[0003] A direct current relay refers to a device that transmits a mechanical drive or current signal using the principle of an electromagnet. The direct current relay is also called an electromagnetic switch, and is generally classified as an electrical circuit switching device.
[0004] The direct current relay includes a fixed contact and a movable contact. The fixed contact is energizably connected to an external power source and load. The fixed contact and the movable contact may be in contact with each other or spaced apart from each other.
[0005] By contact and separation between the fixed contact and the movable contact, energization through the direct current relay is allowed or blocked. The movement is achieved by a driving unit that applies a driving force to the movable contact.
[0006] When the fixed contact and the movable contact are spaced apart, an arc is generated between the fixed contact and the movable contact. An arc is a flow of high-voltage, high-temperature current. Therefore, the generated arc must be quickly discharged from the direct current relay through a predetermined path.
[0007] The discharge path of the arc is formed by a magnet provided in the direct current relay. The magnet forms a magnetic field inside a space where the fixed contact and the movable contact are in contact with each other. The discharge path of the arc may be formed by the electromagnetic force generated by the formed magnetic field and current flow.
[0008] In a conventional direct current relay, the electromagnetic force acting on some fixed contacts is formed toward the inside, that is, toward the central part of the movable contacts. Therefore, the arc generated at the corresponding location cannot be immediately discharged to the outside.
[0009] In the central part of the direct current relay, that is, the space between each fixed contact, several members are provided to drive the movable contact in the up-down direction. For example, a shaft, a spring member inserted through the shaft, and the like are provided at the above position.
[0010] Therefore, if the arc generated is moved toward the central part, and if the arc moved to the central part is not immediately moved to the outside, there is a concern that several members provided at the position may be damaged by the energy of the arc.
[0011] In addition, the direction of the electromagnetic force formed inside a conventional direct current relay depends on the direction of the electric current flowing through the fixed contact. That is, the position of the electromagnetic force formed in the inward direction among the electromagnetic forces generated at each fixed contact differs depending on the direction of the electric current.
[0012] That is, the user must consider the direction of the electric current whenever using the direct

current relay. This may cause inconvenience in use of the direct current relay. In addition, regardless of the user's intention, a situation in which the direction of the electric current applied to the direct current relay is changed due to inexperienced operation cannot be excluded.

[0013] In this case, members provided in the central part of the direct current relay may be damaged by the generated arc. Accordingly, the durability period of the direct current relay may be reduced, and safety accidents may occur.

[0014] Korean Patent Registration No. 10-1696952 discloses a direct current relay. Specifically, it discloses a direct current relay with a structure capable of preventing movement of a movable contact by using a plurality of permanent magnets.

[0015] However, although this type of direct current relay can prevent movement of the movable contact by using a plurality of permanent magnets, there is a limitation in that there is no consideration for a method for controlling the direction of an arc discharge path.

[0016] Korean Patent Registration No. 10-1216824 discloses a direct current relay. Specifically, it discloses a direct current relay with a structure capable of preventing any separation between a movable contact and a fixed contact by using a damping magnet.

[0017] However, this type of direct current relay only proposes a method for maintaining the contact state between the movable contact and the fixed contact. That is, there is a limitation in that a method for forming a discharge path of an arc generated when the movable contact and the fixed contact are separated is not proposed. [0018] (Patent Document 1) Korean Patent Registration No. 10-1696952 (Jan. 16, 2017) [0019] (Patent Document 2) Korean Patent Registration No. 10-1216824 (Dec. 28, 2012)

SUMMARY

[0020] The present disclosure is directed to providing an arc path formation unit capable of quickly extinguishing and discharging an arc generated as the energized electric current is cut off, and a direct current relay including the same.

[0021] The present disclosure is also directed to providing an arc path formation unit that can strengthen the magnitude of force for guiding a generated arc and a direct current relay including the same.

[0022] The present disclosure is also directed to providing an arc path formation unit that can prevent damage to components for energizing due to a generated arc, and a direct current relay including the same.

[0023] The present disclosure is also directed to providing an arc path formation unit through which arcs generated at a plurality of positions can proceed without meeting each other, and a direct current relay including the same.

[0024] The present disclosure is also directed to providing an arc path formation unit and a direct current relay including the same that can achieve the above-described objects without excessive design changes.

[0025] In order to achieve the above objects, the arc path formation unit according to an embodiment of the present disclosure includes an arc chamber in which a plurality of fixed contacts and movable contacts are accommodated; a magnet holder unit disposed outside the arc chamber and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit includes: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of

the second holder facing the second magnet or the other end facing the first magnet, and wherein the first magnet and the second magnet are arranged to be offset from the third magnet and the fourth magnet, respectively, without facing each other, with respect to the central point of the plurality of fixed contacts.
[0026] In addition, a shortest path between the first magnet and the third magnet may overlap the central point of the plurality of fixed contacts in a movement direction of the movable contact, and a shortest path between the second magnet and the fourth magnet may overlap the central point of the plurality of fixed contacts in a movement direction of the movable contact.
[0027] In addition, the first magnet may extend in a direction parallel to an extension direction of the third magnet, and the second magnet may extend in a direction parallel to an extension direction of the fourth magnet.
[0028] In addition, the first magnet and the second magnet may have their respective extension directions intersecting each other.
[0029] In addition, the first magnet may be arranged to be offset from the second magnet without facing each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet may be arranged to be offset from the fourth magnet without facing each other, with the virtual line interposed therebetween.
[0030] In addition, in the magnet unit, a shortest distance between the first magnet and the second magnet may be formed to be equal to a shortest distance between the third magnet and the fourth magnet.
[0031] In addition, the first magnet may be arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet may be arranged to face the fourth magnet each other with the virtual line interposed therebetween.
[0032] In addition, in the magnet unit, a shortest distance between the first magnet and the second magnet may be formed to be equal to a shortest distance between the third magnet and the fourth magnet.
[0033] In addition, the first magnet, the second magnet, the third magnet, and the fourth magnet may be all magnetized to the same polarity.
[0034] In addition, the first magnet and the second magnet may be magnetized to any one polarity of the N pole and the S pole, and the third magnet and the fourth magnet may be magnetized to the other one polarity of the N pole and the S pole.
[0035] In addition, the arc path formation unit according to another embodiment of the present disclosure includes: an arc chamber in which a plurality of fixed contacts and movable contacts are accommodated; a magnet holder unit disposed outside the arc chamber and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit includes: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein at least two of the first magnet, the second magnet, the third magnet, and the fourth magnet are formed in different sizes.
[0036] In addition, the first magnet and the third magnet may be formed in different sizes, and the second magnet and the fourth magnet may be also formed in different sizes.

[0037] In addition, the first magnet and the second magnet may be formed to have a first length in the longitudinal direction, and the third magnet and the fourth magnet may be formed to have a second length in the longitudinal direction.
[0038] In addition, the first magnet may be arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet may be arranged to face the fourth magnet each other with the virtual line interposed therebetween.
[0039] In addition, the first magnet and the second magnet may be formed in different sizes, and the third magnet and the fourth magnet may be also formed in different sizes.
[0040] In addition, the first magnet and the third magnet may be formed to have a first length in the longitudinal direction, and the second magnet and the fourth magnet may be formed to have a second length in the longitudinal direction.
[0041] In addition, the first magnet may be symmetrical to the third magnet with respect to a central point of the plurality of fixed contacts, and the second magnet may be symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts.
[0042] In addition, the first magnet may be formed to have a first length in the longitudinal direction, and the second magnet, the third magnet, and the fourth magnet may be formed to have a second length in the longitudinal direction.
[0043] In addition, the second magnet may be symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts.
[0044] In addition, the third magnet may be arranged to face the fourth magnet each other with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween.
[0045] In addition, the first magnet, the second magnet, the third magnet, and the fourth magnet may be all magnetized to the same polarity.
[0046] In addition, the first magnet and the second magnet may be magnetized to any one polarity of the N pole and the S pole, and the third magnet and the fourth magnet may be magnetized to the other one polarity of the N pole and the S pole.
[0047] In addition, the present disclosure provides an embodiment of a direct current relay, including: a plurality of fixed contacts spaced apart from each other in one direction; a movable contact that is in contact with or spaced apart from the fixed contact; an arc chamber in which a space is formed to accommodate the fixed contact and the movable contact;
a frame surrounding the arc chamber; a magnet holder unit disposed between the outside of the arc chamber and the inside of the frame and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit includes: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein the first magnet and the second magnet are arranged to be offset from the third magnet and the fourth magnet, respectively, without facing each other, with respect to the central point of the plurality of fixed contacts.
[0048] In addition, the first magnet may extend in a direction parallel to an extension direction of the third magnet, and the second magnet may extend in a direction parallel to an extension direction of the fourth magnet, and its extension direction may intersect the extension direction of the first magnet.

[0049] In addition, the first magnet may be arranged to be offset from the second magnet without facing each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet may be arranged to be offset from the fourth magnet without facing each other, with the virtual line interposed therebetween.
[0050] In addition, the first magnet may be arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet may be arranged to face the fourth magnet each other with the virtual line interposed therebetween.
[0051] In addition, the direct current relay according to another embodiment of the present disclosure includes: a plurality of fixed contacts spaced apart from each other in one direction; a movable contact that is in contact with or spaced apart from the fixed contact; an arc chamber in which a space is formed to accommodate the fixed contact and the movable contact;
a frame surrounding the arc chamber; a magnet holder unit disposed between the outside of the arc chamber and the inside of the frame and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit includes: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein at least two of the first magnet, the second magnet, the third magnet, and the fourth magnet are formed in different sizes.
[0052] In addition, the first magnet and the second magnet may be arranged to face each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and are formed to have a first length in the longitudinal direction, and the third magnet and the fourth magnet may be arranged to face each other, with the virtual line interposed therebetween, and be formed to have a second length in the longitudinal direction.
[0053] In addition, the first magnet may be symmetrical to the third magnet with respect to a central point of the plurality of fixed contacts, the second magnet may be symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts, the first magnet and the third magnet may be formed to have a first length in the longitudinal direction, and the second magnet and the fourth magnet may be formed to have a second length in the longitudinal direction.
[0054] In addition, the first magnet may be formed to have a first length in the longitudinal direction, and the second magnet, the third magnet, and the fourth magnet may be formed to have a second length in the longitudinal direction.
[0055] Among the various effects of the present disclosure, the effects that can be obtained through the above-described solution are as follows.
[0056] First, the arc path formation unit includes a magnet unit. The magnet unit forms a magnetic field inside the arc path formation unit. The formed magnetic field forms an electromagnetic force together with an electric current energized in the fixed contact and the movable contact accommodated in the arc path formation unit.
[0057] In this case, the generated arc is formed in a direction away from each fixed contact. An arc generated when the fixed contact and the movable contact are separated may be guided by the electromagnetic force.
[0058] Therefore, the generated arc can be quickly extinguished and discharged to the outside of the arc path formation unit and the direct current relay.

[0059] In addition, the magnet unit may include a plurality of magnets. The plurality of magnets is formed to strengthen the intensity of electromagnetic force formed in the vicinity of each fixed contact. That is, the arc path formation units formed near the same fixed contact by different magnets are formed in the same direction.
[0060] Therefore, the intensity of the magnetic field formed near each fixed contact and the intensity of the electromagnetic force dependent on the intensity of the magnetic field can also be strengthened. As a result, the intensity of the electromagnetic force guiding the generated arc is enhanced, so that the generated arc can be effectively extinguished and discharged.
[0061] In addition, the direction of the electromagnetic force formed by the magnetic field formed by the magnet unit and the electric current energized in the fixed contact and the movable contact is formed in a direction away from the central portion.
[0062] Furthermore, since the intensity of the magnetic field and electromagnetic force is strengthened by the magnet unit as described above, the generated arc can be extinguished and moved quickly in a direction away from the central portion.
[0063] Therefore, damage to various components provided in the vicinity of the central portion for the operation of the direct current relay can be prevented.
[0064] In addition, in various embodiments, a plurality of fixed contacts can be provided. The magnet unit provided in the arc path formation unit forms magnetic fields in different directions in the vicinity of each fixed contact. Therefore, paths of arcs generated in the vicinity of each fixed contact proceed in different directions.
[0065] Therefore, the arcs generated in the vicinity of each fixed contact do not meet each other. Accordingly, malfunctions or safety accidents or the like that may occur due to collisions of arcs generated at different locations can be prevented.
[0066] In addition, the magnet unit and the magnet holder unit are located inside the frame surrounding the arc chamber. That is, the magnet unit and the magnet holder unit are located between the inside of the frame and the outside of the arc chamber.
[0067] Therefore, no separate design change is required to place the magnet unit and the magnet holder unit outside the arc chamber.
[0068] Therefore, the arc path formation unit according to various embodiments of the present disclosure can be provided in the direct current relay without excessive design change. Furthermore, time and cost for applying the arc path formation unit according to various embodiments of the present disclosure can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0069] FIG. **1** is a front cross-sectional view illustrating a direct current relay according to an exemplary embodiment of the present disclosure.
[0070] FIG. **2** is a plan cross-sectional view illustrating the direct current relay of FIG. **1**.
[0071] FIG. **3** is a conceptual diagram illustrating an arc path formation unit according to a first embodiment of the present disclosure.
[0072] FIGS. **4** to **5** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **3**.
[0073] FIG. **6** is a conceptual diagram illustrating another example of a magnet unit provided in the arc path formation unit of FIG. **3**.
[0074] FIGS. **7** to **8** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **6**.
[0075] FIG. **9** is a conceptual diagram illustrating yet another example of a magnet unit provided in the arc path formation unit of FIG. **3**.

[0076] FIGS. **10** to **11** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **9**.

[0077] FIG. **12** is a conceptual diagram illustrating an arc path formation unit according to a second embodiment of the present disclosure.

[0078] FIGS. **13** to **14** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **12**.

[0079] FIG. **15** is a conceptual diagram illustrating another example of a magnet unit provided in the arc path formation unit of FIG. **12**.

[0080] FIGS. **16** to **17** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **15**.

[0081] FIG. **18** is a conceptual diagram illustrating yet another example of a magnet unit provided in the arc path formation unit of FIG. **12**.

[0082] FIG. **19** is a conceptual diagram illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **18**.

[0083] FIG. **20** is a conceptual diagram illustrating an arc path formation unit according to a third embodiment of the present disclosure.

[0084] FIGS. **21** to **22** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **20**.

[0085] FIG. **23** is a conceptual diagram illustrating another example of a magnet unit provided in the arc path formation unit of FIG. **20**.

[0086] FIGS. **24** to **25** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **23**.

[0087] FIG. **26** is a conceptual diagram illustrating yet another example of a magnet unit provided in the arc path formation unit of FIG. **20**.

[0088] FIGS. **27** to **28** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **26**.

[0089] FIG. **29** is a conceptual diagram illustrating yet another example of a magnet unit provided in the arc path formation unit of FIG. **20**.

[0090] FIGS. **30** to **31** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **29**.

[0091] FIG. **32** is a conceptual diagram illustrating an arc path formation unit according to a fourth embodiment of the present disclosure.

[0092] FIGS. **33** to **34** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **32**.

[0093] FIG. **35** is a conceptual diagram illustrating another example of a magnet unit provided in the arc path formation unit of FIG. **32**.

[0094] FIGS. **36** to **37** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **35**.

[0095] FIG. **38** is a conceptual diagram illustrating yet another example of a magnet unit provided in the arc path formation unit of FIG. **32**.

[0096] FIGS. **39** to **40** are conceptual diagrams illustrating the magnetic field and arc path formed by the arc path formation unit of FIG. **38**.

DETAILED DESCRIPTION

[0097] Hereinafter, the arc path formation units **100**, **200**, **300**, and **400** and the direct current relay **1** including the same according to embodiments of the present disclosure will be described in more detail with reference to the drawings.

[0098] In the following description, in order to clarify the features of the present disclosure, descriptions of some components may be omitted.

[0099] In this specification, even in different embodiments, the same reference numerals will designate the same elements, and a redundant description thereof will be omitted.

[0100] The accompanying drawings are only for easy understanding of the embodiments disclosed herein, and the technical ideas disclosed herein are not limited by the accompanying drawings.
[0101] Expressions in the singular include plural expressions unless the context clearly indicates otherwise.

1. Description of the Direct Current Relay **1** According to an Embodiment of the Present Disclosure

[0102] Hereinafter, a direct current relay **1** according to an embodiment of the present disclosure will be described with reference to FIGS. **1** to **2**.
[0103] The direct current relay **1** according to an embodiment of the present disclosure includes a frame unit **10**, a switch unit **20**, a core unit **30**, and a movable contact unit **40**. In addition, the direct current relay **1** includes an arc path formation unit **100**, **200**, **300**, **400**.
[0104] The arc path formation unit **100**, **200**, **300**, **400** may form a discharge path of the generated arc.
[0105] Hereinafter, the configuration of the direct current relay **1** according to an embodiment of the present disclosure will be described with reference to the attached drawings, but the frame unit **10**, switch unit **20**, the core unit **30**, the movable contact unit **40**, and the arc path formation unit **100**, **200**, **300**, **400** will be described in separate sections.
[0106] The arc path formation unit **100**, **200**, **300**, **400** according to various embodiments described below will be described on the premise that they are provided in a direct current relay **1**. However, it will be understood that the arc path formation unit **100**, **200**, **300**, **400** may be applied to a type of device, such as electromagnetic contactor and electromagnetic switch, capable of energizing electric current or blocking electric current with the outside by contact and separation between the fixed contact and the movable contact.

(1) Description of the Frame Unit **10**

[0107] The frame unit **10** forms the outer side of the direct current relay **1**. A predetermined space is formed inside the frame unit **10**. Various devices performing a function of applying or blocking electric current transferred from the outside by the direct current relay **1** may be accommodated in the space. That is, the frame unit **10** functions as a kind of housing **41**.
[0108] In an embodiment, the frame unit **10** is formed of an insulating material such as synthetic resin, so that arbitrary energization of the inside and outside of the frame unit **10** can be prevented.
[0109] In the illustrated embodiment, the frame unit **10** includes an upper frame **11**, a lower frame **12**, an insulating plate **13**, and a support plate **14**.
[0110] The upper frame **11** forms an upper side of the frame unit **10**. A predetermined space is formed inside the upper frame **11**.
[0111] The switch unit **20** and the movable contact unit **40** may be accommodated in the inner space of the upper frame **11**. In addition, the arc path formation unit **100**, **200**, **300**, **400** may be accommodated in the inner space of the upper frame **11**.
[0112] A fixed contact **22** of the switch unit **20** is positioned on one side of the upper frame **11**, that is, on the upper side in the illustrated embodiment. A portion of the fixed contact **22** may be exposed on the upper side of the upper frame **11**, and may be energizably connected to an external power source or load. To this end, a through hole through which the fixed contact **22** is through-coupled may be formed on one side of the upper frame **11**.
[0113] The lower frame **12** forms a lower side of the frame unit **10**. A predetermined space is formed inside the lower frame **12**. The core unit **30** may be accommodated in the inner space of the lower frame **12**.
[0114] The lower frame **12** may be coupled to the upper frame **11**. The insulating plate **13** and the support plate **14** may be provided in a space between the lower frame **12** and the upper frame **11**.
[0115] The insulating plate **13** is positioned between the upper frame **11** and the lower frame **12**.
[0116] The insulating plate **13** electrically separates the upper frame **11** and the lower frame **12**. To this end, the insulating plate **13** is preferably formed of an insulating material such as synthetic

resin.
[0117] By the insulating plate **13**, any energization between the switch unit **20**, the movable contact unit **40**, and the arc path formation unit **100**, **200**, **300**, **400** accommodated in the upper frame **11** and the core unit **30** accommodated in the lower frame **12** may be prevented.
[0118] A through hole (not shown) is formed in the central portion of the insulating plate **13**. A shaft **44** of the movable contact unit **40** is through-coupled to the through hole so as to be movable in the up-down direction.
[0119] The support plate **14** is positioned below the insulating plate **13**.
[0120] The support plate **14** supports the lower side of the insulating plate **13**.
[0121] The support plate **14** is positioned between the upper frame **11** and the lower frame **12**.
[0122] The support plate **14** physically separates the upper frame **11** and the lower frame **12**.
[0123] The support plate **14** may be formed of a magnetic material. Therefore, the support plate **14** may form a magnetic circuit together with a yoke **33**. A driving force for moving a movable core **32** of the core unit **30** toward a stationary core **31** may be formed by the magnetic circuit.
[0124] A through hole (not shown) is formed in the central portion of the support plate **14**. The shaft **44** is through-coupled to the through hole so as to be movable in the up-down direction.
[0125] Therefore, when the movable core **32** is moved in a direction toward the stationary core **31** or away from the stationary core **31**, the shaft **44** and a movable contact **43** connected to the shaft **44** may also be moved together in the same direction.
(2) Description of the Switch Unit **20**
[0126] The switch unit **20** allows or blocks electric current energization according to the operation of the core unit **30**. Specifically, the switch unit **20** may allow or block energizing electric current by contacting or separating the fixed contact **22** and the movable contact **43**.
[0127] The switch unit **20** is accommodated in the inner space of the upper frame **11**. The switch unit **20** may be electrically and physically separated from the core unit **30** by the insulating plate **13** and the support plate **14**.
[0128] In the illustrated embodiment, the switch unit **20** includes an arc chamber **21**, a fixed contact **22**, and a sealing member **23**.
[0129] The arc chamber **21** extinguishes an arc generated when the fixed contact **22** and the movable contact **43** are separated from each other in the inner space. Accordingly, the arc chamber **21** may be referred to as an “arc extinguishing unit.”
[0130] The arc chamber **21** hermetically accommodates the fixed contact **22** and the movable contact **43**. That is, the fixed contact **22** and the movable contact **43** are accommodated inside the arc chamber **21**. Therefore, an arc generated when the fixed contact **22** and the movable contact **43** are separated from each other does not arbitrarily leak to the outside.
[0131] A gas for extinguishing may be filled in the arc chamber **21**. The gas for extinguishing allows the generated arc to be extinguished and discharged to the outside of the direct current relay **1** through a preset path. To this end, a communication hole (not shown) may be formed through a wall surrounding the inner space of the arc chamber **21**.
[0132] In an embodiment, the arc chamber **21** may be formed of an insulating material. In another embodiment, the arc chamber **21** may be formed of a material having high pressure resistance and high heat resistance. This is due to the generated arc being the flow of high temperature and high pressure electrons. For example, the arc chamber **21** may be formed of a ceramic material.
[0133] A plurality of through holes may be formed on the upper side of the arc chamber **21**. A fixed contact **22** is through-coupled to each of the through holes.
[0134] In the illustrated embodiment, two fixed contacts **22** are provided, including a first fixed contact **22***a* and a second fixed contact **22***b*. Accordingly, two through holes may also be formed on the upper side of the arc chamber **21**.
[0135] When the fixed contact **22** is through-coupled to the through hole, the through hole is sealed. That is, the fixed contact **22** is hermetically coupled to the through hole. Accordingly, the

generated arc is not discharged to the outside through the through hole.
[0136] The lower side of the arc chamber **21** may be open. The insulating plate **13** and the sealing member **23** are in contact with the lower side of the arc chamber **21**. That is, the lower side of the arc chamber **21** is sealed by the insulating plate **13** and the sealing member **23**.
[0137] Accordingly, the arc chamber **21** may be electrically and physically separated from the outer space of the upper frame **11**.
[0138] The arc extinguished in the arc chamber **21** is discharged to the outside of the direct current relay **1** through a preset path. In an embodiment, the extinguished arc may be discharged to the outside of the arc chamber **21** through the communication hole.
[0139] An arc path formation unit **100**, **200**, **300**, **400** may be provided on the outside of the arc chamber **21**. The arc path formation unit **100**, **200**, **300**, **400** may form a magnetic field for forming an arc path A.P for an arc generated inside the arc chamber **21**. This will be described later in detail.
[0140] The fixed contact **22** is in contact with or separated from the movable contact **43** to apply or block internal and external energization of the direct current relay **1**.
[0141] Specifically, when the fixed contact **22** is in contact with the movable contact **43**, the inside and outside of the direct current relay **1** may be energized. On the other hand, when the fixed contact **22** is separated from the movable contact **43**, the energization of the inside and outside of the direct current relay **1** is blocked.
[0142] As can be seen from the name, the fixed contact **22** is not moved. That is, the fixed contact **22** is fixedly coupled to the upper frame **11** and the arc chamber **21**. Thus, the contact and separation of the fixed contact **22** and the movable contact **43** are achieved by the movement of the movable contact **43**.
[0143] One end of the fixed contact **22**, that is, the upper end in the illustrated embodiment, is exposed to the outside of the upper frame **11**. A power source or load is energizably connected to the one end, respectively.
[0144] A plurality of fixed contacts **22** may be provided. In the illustrated embodiment, two fixed contacts **22** are provided, including a first fixed contact **22***a* on the left side and a second fixed contact **22***b* on the right side.
[0145] The first fixed contact **22***a* is positioned to be biased to one side, that is, to the left in the illustrated embodiment, from the center in the longitudinal direction of the movable contact **43**. In addition, the second fixed contact **22***b* is positioned to be biased to the other side, that is, to the right in the illustrated embodiment, from the center in the longitudinal direction of the movable contact **43**.
[0146] A power source may be energizably connected to any one of the first fixed contact **22***a* and the second fixed contact **22***b*. In addition, a load may be energizably connected to the other one of the first fixed contact **22***a* and the second fixed contact **22***b*.
[0147] The direct current relay **1** according to an embodiment of the present disclosure may form an arc path A.P regardless of the direction of a power source or load connected to the fixed contact **22**. This is achieved by the arc path formation unit **100**, **200**, **300**, **400**, a detailed description of which will be provided later.
[0148] The other end of the fixed contact **22**, that is, the lower end in the illustrated embodiment, extends toward the movable contact **43**.
[0149] When the movable contact **43** is moved in a direction toward the fixed contact **22**, that is, upward in the illustrated embodiment, the lower end comes into contact with the movable contact **43**. Accordingly, the outside and the inside of the direct current relay **1** may be energized.
[0150] The lower end of the fixed contact **22** is located inside the arc chamber **21**.
[0151] When the control power is cut off, the movable contact **43** is spaced apart from the fixed contact **22** by an elastic force of a return spring **36**.
[0152] In this case, as the fixed contact **22** and the movable contact **43** are spaced apart from each other, an arc is generated between the fixed contact **22** and the movable contact **43**. The generated

arc may be extinguished by a gas for extinguishing inside the arc chamber **21** and may be discharged to the outside along a path formed by the arc path formation unit **100**, **200**, **300**, **400**.

[0153] The sealing member **23** blocks any communication between the arc chamber **21** and a space inside the upper frame **11**.

[0154] The sealing member **23** seals the lower side of the arc chamber **21** together with the insulating plate **13** and the support plate **14**. Specifically, the upper side of the sealing member **23** is coupled to the lower side of the arc chamber **21**. In addition, the radially inner side of the sealing member **23** is coupled to the outer circumference of the insulating plate **13**, and the lower side of the sealing member **23** is coupled to the support plate **14**.

[0155] Therefore, the arc generated in the arc chamber **21** and the arc extinguished by the gas for extinguishing do not arbitrarily leak into the inner space of the upper frame **11**.

[0156] In addition, the sealing member **23** may be configured to block any communication between the inner space of the cylinder **37** and the inner space of the frame unit **10**.

(3) Description of the Core Unit **30**

[0157] The core unit **30** moves the movable contact unit **40** upward according to the application of the control power. In addition, when the application of the control power is released, the core unit **30** moves the movable contact unit **40** downward again.

[0158] The core unit **30** may be energizably connected to an external control power (not shown) to receive the control power.

[0159] The core unit **30** is located below the switch unit **20**. In addition, the core unit **30** is accommodated inside the lower frame **12**. The core unit **30** and the switch unit **20** may be electrically and physically separated from each other by the insulating plate **13** and the support plate **14**.

[0160] The movable contact unit **40** is positioned between the core unit **30** and the switch unit **20**. The movable contact unit **40** may be moved by the driving force applied by the core unit **30**. Accordingly, the movable contact **43** and the fixed contact **22** may be brought into contact with each other so that the direct current relay **1** may be energized.

[0161] In the illustrated embodiment, the core unit **30** includes a stationary core **31**, a movable core **32**, a yoke **33**, a bobbin **34**, a coil **35**, a return spring **36**, and a cylinder **37**.

[0162] The stationary core **31** is magnetized by a magnetic field generated in the coil **35** to generate an electromagnetic repulsive force. Due to the electromagnetic repulsive force, the movable core **32** is moved in a direction away from the stationary core **31**.

[0163] The stationary core **31** is not moved. That is, the stationary core **31** is fixedly coupled to the support plate **14** and the cylinder **37**.

[0164] The stationary core **31** may be provided in any form capable of generating electromagnetic force by being magnetized by a magnetic field. In an embodiment, the stationary core **31** may be provided as a permanent magnet or an electromagnet or the like.

[0165] The stationary core **31** partially accommodates the lower side of the cylinder **37**. In addition, the inner circumference of the stationary core **31** is in contact with the outer circumference of the cylinder **37**.

[0166] A through hole (not shown) is formed at the central portion of the stationary core **31**. The shaft **44** is through-coupled to the through hole so as to be movable up and down.

[0167] When the control power is applied, the movable core **32** is moved in a direction away from the stationary core **31** by electromagnetic repulsive force generated by the stationary core **31**.

[0168] As the movable core **32** is moved, the shaft **44** coupled to the movable core **32** is moved in a direction away from the stationary core **31**, that is, upward in the illustrated embodiment. In addition, as the shaft **44** is moved, the movable contact unit **40** coupled to the shaft **44** is also moved upward.

[0169] Accordingly, the fixed contact **22** and the movable contact **43** may be brought into contact with each other so that the direct current relay **1** may be energized with an external power source or

load.
[0170] The movable core **32** may be provided in any shape capable of being subjected to repulsive force by electromagnetic force. In an embodiment, the movable core **32** may be formed of a magnetic material or may be provided as a permanent magnet or an electromagnet or the like.
[0171] The movable core **32** is accommodated inside the cylinder. In addition, the movable core **32** may be moved in the longitudinal direction of the cylinder **37** inside the cylinder **37**, that is, in the up-down direction in the illustrated embodiment.
[0172] Specifically, the movable core **32** may be moved in a direction toward the stationary core **31** and in a direction away from the stationary core **31**.
[0173] The movable core **32** is coupled to the shaft **44**. The movable core **32** may be moved integrally with the shaft **44**. When the movable core **32** is moved upward or downward, the shaft **44** is also moved upward or downward. Accordingly, the movable contact **43** is also moved upward or downward.
[0174] The movable core **32** is located above the stationary core **31**. The movable core **32** may be spaced apart from the stationary core **31** by a predetermined distance. The predetermined distance may be defined as a distance at which the movable core **32** can be moved in the up-down direction.
[0175] The movable core **32** is formed extending in the longitudinal direction. Inside the movable core **32**, a hollow portion extending in the longitudinal direction is formed recessed by a predetermined distance. The return spring **36** and the lower side of the shaft **44** coupled through the return spring **36** are partially accommodated in the hollow portion.
[0176] A through hole is formed through the lower side of the hollow portion in the longitudinal direction. The hollow portion and the through hole communicate with each other. A lower end of the shaft **44** inserted into the hollow portion may progress toward the through hole.
[0177] At the lower end of the movable core **32**, a space portion is formed recessed by a predetermined distance. The space portion communicates with the through hole. The lower head portion of the shaft **44** is located in the space portion.
[0178] The yoke **33** forms a magnetic circuit as control power is applied. The magnetic circuit formed by the yoke **33** may be configured to control the direction of the magnetic field formed by the coil **35**.
[0179] Accordingly, when the control power is applied, the coil **35** may generate a magnetic field so that the movable core **32** is moved in a direction away from the stationary core **31**.
[0180] In an embodiment, the yoke **33** may be formed of a conductive material capable of energizing.
[0181] The yoke **33** is accommodated inside the lower frame **12**. The yoke **33** surrounds the coil **35**. The coil **35** may be accommodated inside the yoke **33** to be spaced apart from the inner circumferential surface of the yoke **33** by a predetermined distance. The bobbin **34** is accommodated inside the yoke **33**. That is, the yoke **33**, the coil **35**, and the bobbin **34** around which the coil **35** is wound are sequentially arranged in a radially inward direction from the outer circumference of the lower frame **12**.
[0182] The upper side of the yoke **33** is in contact with the support plate **14**. In addition, the outer circumference of the yoke **33** may be positioned to contact the inner circumference of the lower frame **12** or to be spaced apart from the inner circumference of the lower frame **12** by a predetermined distance.
[0183] The coil **35** is wound on the bobbin **34**.
[0184] The bobbin **34** is accommodated inside the yoke **33**.
[0185] The bobbin **34** may include a flat plate-shaped upper portion and a flat plate-shaped lower portion, and a cylindrical pillar portion extending in a longitudinal direction and connecting the upper portion and the lower portion. That is, the bobbin **34** has a bobbin shape.
[0186] An upper portion of the bobbin **34** is in contact with a lower side of the support plate **14**. The coil **35** is wound around the pillar portion of the bobbin **34**. The thickness of the coil **35** to be

wound may be equal to or smaller than the diameters of the upper and lower portions of the bobbin **34**.
[0187] A hollow portion extending in the longitudinal direction is formed through the pillar portion of the bobbin **34**. The cylinder **37** may be accommodated in the hollow portion. The pillar portion of the bobbin **34** may be arranged to have the same central axis as the stationary core **31**, the movable core **32**, and the shaft **44**.
[0188] The coil **35** generates a magnetic field by an applied control power. The stationary core **31** may be magnetized by a magnetic field generated by the coil **35**, and an electromagnetic repulsive force may be applied to the movable core **32**.
[0189] The coil **35** is wound around the bobbin **34**. Specifically, the coil **35** is wound around the pillar portion of the bobbin **34** and stacked radially outward of the pillar portion. The coil **35** is accommodated inside the yoke **33**.
[0190] When the control power is applied, the coil **35** generates a magnetic field. In this case, the intensity or direction or the like of the magnetic field generated by the coil **35** may be controlled by the yoke **33**. The stationary core **31** may be magnetized by the magnetic field generated by the coil **35**.
[0191] When the stationary core **31** is magnetized, the movable core **32** is subjected to an electromagnetic force, that is, a repulsive force, in a direction away from the stationary core **31**. Accordingly, the movable core **32** is moved in a direction toward the stationary core **31**, that is, upward in the illustrated embodiment.
[0192] The return spring **36** provides a restoring force for returning the movable core **32** to its original position when application of the control power is released after the movable core **32** is moved in a direction away from the stationary core **31**.
[0193] As the movable core **32** is moved toward the stationary core **31** the return spring **36** is compressed and stores a restoring force. At this time, the stored restoring force is preferably smaller than the electromagnetic repulsive force applied to the movable core **32** after the stationary core **31** is magnetized. This is to prevent the movable core **32** from being arbitrarily returned to its original position by the return spring **36** while the control power is applied.
[0194] When the application of the control power is released, the movable core **32** receives a restoring force by the return spring **36**. Of course, gravity due to the empty weight of the movable core **32** may also act on the movable core **32**. Accordingly, the movable core **32** may be moved in a direction away from the stationary core **31** and returned to its original position.
[0195] The return spring **36** may be provided in any shape capable of deforming, storing restoring force, returning to its original shape, and transmitting the restoring force to the outside. In an embodiment, the return spring **36** may be provided as a coil spring **35**.
[0196] The shaft **44** is coupled through the return spring **36**. The shaft **44** may be moved in the up-down direction regardless of the shape deformation of the return spring **36** in a state in which the return spring **36** is coupled.
[0197] The return spring **36** is accommodated in a hollow portion formed recessed on the upper side of the movable core **32**.
[0198] The cylinder **37** accommodates the movable core **32**, the return spring **36**, and the shaft **44**. The movable core **32** and the shaft **44** may be moved inside the cylinder **37** in upward and downward directions.
[0199] The cylinder **37** is located in the hollow portion formed in the pillar portion of the bobbin **34**. The side surface of the cylinder **37** is in contact with the inner circumferential surface of the pillar portion of the bobbin **34**.
[0200] The upper end of the cylinder **37** is in contact with the lower surface of the support plate **14**. The lower surface of the cylinder **37** may be in contact with the stationary core **31**.
(4) Description of the Movable Contact Unit **40**
[0201] The movable contact unit **40** includes a movable contact **43** and a component for moving

the movable contact **43**. The direct current relay **1** may be energized with an external power source or load by the movable contact unit **40**.
[0202] The movable contact unit **40** is accommodated in the inner space of the upper frame **11**. In addition, the movable contact unit **40** is accommodated in the arc chamber **21** to be movable up and down.
[0203] A fixed contact **22** is positioned above the movable contact unit **40**. The movable contact unit **40** is accommodated in the arc chamber **21** to be movable in a direction toward the fixed contact **22** and in a direction away from the fixed contact **22**.
[0204] A core unit **30** is positioned below the movable contact unit **40**. The movement of the movable contact unit **40** may be achieved by the movement of the movable core **32**.
[0205] In the illustrated embodiment, the movable contact unit **40** includes a housing **41**, a cover **42**, a movable contact **43**, a shaft **44**, and an elastic portion **45**.
[0206] The housing **41** accommodates the movable contact **43** and the elastic portion **45** elastically supporting the movable contact **43**.
[0207] In the illustrated embodiment, the housing **41** is open on one side and the other side opposite thereto. The movable contact **43** may be inserted through the open portion. The non-open side of the housing **41** may be configured to surround the accommodated movable contact **43**.
[0208] The cover **42** is provided above the housing **41**.
[0209] The cover **42** covers the upper surface of the movable contact **43** accommodated in the housing **41**.
[0210] The housing **41** and the cover **42** are preferably formed of an insulating material to prevent unintentional energization. In an embodiment, the housing **41** and the cover **42** may be formed of a synthetic resin or the like.
[0211] The lower side of the housing **41** is connected to the shaft **44**. When the movable core **32** connected to the shaft **44** is moved upward or downward, the housing **41** and the movable contact **43** accommodated therein may also be moved upward or downward.
[0212] The housing **41** and the cover **42** may be coupled by any member. In an embodiment, the housing **41** and the cover **42** may be coupled by a fastening member (not shown) such as a bolt or nut.
[0213] The movable contact **43** comes into contact with the fixed contact **22** according to the application of control power, so that the direct current relay **1** is energized with an external power source and load. In addition, the movable contact **43** is separated from the fixed contact **22** when the application of the control power is released, so that the direct current relay **1** is made not energizable with an external power source and load.
[0214] The movable contact **43** is positioned adjacent to the fixed contact **22**.
[0215] The upper side of the movable contact **43** is partially covered by the cover **42**. In an embodiment, a portion of the upper surface of the movable contact **43** may be in contact with the lower surface of the cover **42**.
[0216] The lower side of the movable contact **43** is elastically supported by the elastic portion **45**. To prevent the movable contact **43** from moving arbitrarily downward, the elastic portion **45** may elastically support the movable contact **43** in a compressed state by a predetermined distance.
[0217] The movable contact **43** is formed extending in the longitudinal direction, that is, in the left-right direction in the illustrated embodiment. That is, the length of the movable contact **43** is formed longer than the width. Therefore, both ends in the longitudinal direction of the movable contact **43** accommodated in the housing **41** are exposed to the outside of the housing **41**.
[0218] Contact protrusion portions protruding upward by a predetermined distance may be formed from the both ends. The fixed contact **22** is in contact with the contact protrusion portion.
[0219] The contact protrusion portion may be formed at a position corresponding to each fixed contact **22**. Accordingly, the moving distance of the movable contact **43** may be reduced, and contact reliability between the fixed contact **22** and the movable contact **43** may be improved.

[0220] The width of the movable contact **43** may be the same as the distance at which each side surface of the housing **41** is spaced apart from each other. That is, when the movable contact **43** is accommodated in the housing **41**, both side surfaces of the movable contact **43** in the width direction may contact inner surfaces of each side surface of the housing **41**. Accordingly, the state in which the movable contact **43** is accommodated in the housing **41** can be stably maintained.
[0221] The shaft **44** transmits a driving force generated as the core unit **30** is operated to the movable contact unit **40**. Specifically, the shaft **44** is connected to the movable core **32** and the movable contact **43**. When the movable core **32** is moved upward or downward, the movable contact **43** may be also moved upward or downward by the shaft **44**.
[0222] The shaft **44** is formed extending in the longitudinal direction, that is, in the up-down direction in the illustrated embodiment.
[0223] The lower end of the shaft **44** is inserted into and coupled to the movable core **32**. When the movable core **32** is moved in the up-down direction, the shaft **44** may be moved together with the movable core **32** in the up-down direction.
[0224] The return spring **36** is coupled through the body portion of the shaft **44**.
[0225] The upper end of the shaft **44** is coupled to the housing **41**. When the movable core **32** is moved, the shaft **44** and the housing **41** may be moved together.
[0226] Upper and lower ends of the shaft **44** may be formed to have larger diameters than the body portion of the shaft **44**. Accordingly, the shaft **44** may be stably coupled to the housing **41** and the movable core **32**.
[0227] The elastic portion **45** elastically supports the movable contact **43**. When the movable contact **43** comes into contact with the fixed contact **22**, the movable contact **43** tends to be spaced apart from the fixed contact **22** by electromagnetic repulsive force. In this case, the elastic portion **45** elastically supports the movable contact **43** to prevent the movable contact **43** from being arbitrarily separated from the fixed contact **22**.
[0228] The elastic portion **45** may be provided in any form capable of storing a restoring force by deformation of a shape and providing the stored restoring force to other members. In an embodiment, the elastic portion **45** may be provided as a coil spring **35**.
[0229] One end of the elastic portion **45** facing the movable contact **43** is in contact with the lower side of the movable contact **43**. In addition, the other end facing the one end is in contact with the upper side of the housing **41**.
[0230] The elastic portion **45** may elastically support the movable contact **43** in a state in which a restoring force is stored after being compressed by a predetermined distance. Accordingly, even if an electromagnetic repulsive force is generated between the movable contact **43** and the fixed contact **22**, the movable contact **43** is not moved arbitrarily.
[0231] For stable coupling of the elastic portion **45**, a protrusion portion (not shown) inserted into the elastic portion **45** may protrude from the lower side of the movable contact **43**. Likewise, a protrusion portion (not shown) inserted into the elastic portion **45** may protrude from the upper side of the housing **41**.
2. Description of the Arc Path Formation Unit **100** According to the First Embodiment of the Present Disclosure
[0232] Hereinafter, the arc path formation unit **100** according to the first embodiment of the present disclosure will be described with reference to FIGS. **3** to **11**.
[0233] The arc path formation unit **100** forms a magnetic field inside the arc chamber **21**. An electromagnetic force is formed inside the arc chamber **21** by the electric current energizing through the direct current relay **1** and the formed magnetic field.
[0234] An arc generated as the fixed contact **22** and the movable contact **43** are separated is moved out of the arc chamber **21** by the formed electromagnetic force. Specifically, the generated arc is moved along the direction of the formed electromagnetic force. Accordingly, it can be said that the arc path formation unit **100** forms an arc path A.P, which is a path through which the generated arc

flows.
[0235] The arc path formation unit **100** is located in a space formed inside the upper frame **11**. The arc path formation unit **100** is arranged to surround the arc chamber **21**. In other words, the arc chamber **21** is located inside the arc path formation unit **100**.
[0236] The fixed contact **22** and the movable contact **43** are positioned inside the arc path formation unit **100**. An arc generated when the fixed contact **22** and the movable contact **43** are separated may be guided by the electromagnetic force formed by the arc path formation unit **100**.
[0237] The arc path formation unit **100** according to the present embodiment includes a magnet holder unit **110** and a magnet unit **120**.
[0238] The magnet holder unit **110** forms the skeleton of the arc path formation unit **100** and fixes the magnet unit **120**, which will be described later, to the outside of the arc chamber **21**.
[0239] The magnet holder unit **110** is disposed outside the arc chamber **21** and inside the upper frame **11**.
[0240] The fixed contact **22** and the movable contact **43** are located radially inside the magnet holder unit **110**. The central portion of the fixed contact **22** and the movable contact **43** may be defined as a central portion C. In the illustrated embodiment, the magnet holder unit **110** is arranged so that its center corresponds to the central portion C of the fixed contact **22** and the movable contact **43**.
[0241] The central portion C is located between the first fixed contact **22***a* and the second fixed contact **22***b*. In addition, the central portion of the movable contact unit **40** is positioned vertically below the central portion C. That is, central portions of the housing **41**, the cover **42**, the movable contact **43**, the shaft **44**, the elastic portion **45** or the like are positioned vertically below the central portion C.
[0242] Therefore, when the generated arc is moved toward the central portion C, damage to the above components may occur. To prevent this, the arc path formation unit **100** according to the present embodiment includes a magnet unit **120**. A detailed description of this will be provided later along with a description of the magnet unit **120**.
[0243] In an embodiment, the magnet holder unit **110** may be formed of an electrically conductive material. In the above embodiment, the magnet holder unit **110** may be magnetized to the same polarity as a plurality of adjacent magnets.
[0244] The magnet holder unit **110** may include a plurality of holders. Each holder may be coupled with a plurality of magnets. In an embodiment, a plurality of magnets attached to one holder are all magnetized to the same polarity.
[0245] In the illustrated embodiment, the magnet holder unit **110** includes a total of two holders, including a first holder **111** and a second holder **112**.
[0246] The first holder **111** and the second holder **112** are arranged to be spaced apart from each other. That is, an empty space is formed between the first holder **111** and the second holder **112**. The space may function as a passage through which the arc generated in the arc chamber **21** is discharged.
[0247] In addition, the first holder **111** and the second holder **112** are arranged in a direction parallel to the arrangement direction of a plurality of fixed contacts **22**.
[0248] The first holder **111** and the second holder **112** are each bent and extended at a predetermined angle. In addition, the edges of the bent portions of the first holder **111** and the second holder **112** may be chamfered. In an embodiment, the predetermined angle may be a right angle.
[0249] The first holder **111** and the second holder **112** may be in contact with or fixedly coupled to the inner circumferential surface of the upper frame **11**. Accordingly, the first holder **111** and the second holder **112** are preferably formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0250] The first holder **111** and the second holder **112** are arranged so that the concave portions of

each of the bent portions face each other, with the central portion C of the fixed contact **22** and the movable contact **43** interposed therebetween.
[0251] In addition, the first holder **111** and the second holder **112** are formed in shapes that correspond to each other. In the illustrated embodiment, the first holder **111** and the second holder **112** are formed in a structure that is symmetrical to each other with respect to the central portion C of the plurality of fixed contacts **22** and the movable contact **43**.
[0252] The first holder **111** includes a first outer surface **111***a* and a first inner surface **111***b*.
[0253] The first outer surface **111***a* is located on one surface of the first holder **111** opposite to the fixed contact **22** and the movable contact **43**. In addition, the first outer surface **111***a* is disposed adjacent to the inner circumferential surface of the upper frame **11**. In an embodiment, the first outer surface **111***a* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0254] The first inner surface **111***b* is positioned on the other surface opposite to the first outer surface **111***a* of the first holder **111**. In addition, the first inner surface **111***b* is arranged to face the outer circumferential surface of the arc chamber **21** with the first magnet **121** and the second magnet **122** interposed therebetween. In an embodiment, the first inner surface **111***b* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0255] The first inner surface **111***b* is coupled to the first magnet **121** and the second magnet **122** of the magnet unit **120**, which will be described later.
[0256] The second holder **112** includes a second outer surface **112***a* and a second inner surface **112***b*.
[0257] The second outer surface **112***a* is located on one surface of the second holder **112** opposite to the fixed contact **22** and the movable contact **43**. In addition, the second outer surface **112***a* is disposed adjacent to the inner circumferential surface of the upper frame **11**. In an embodiment, the second outer surface **112***a* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0258] The second inner surface **112***b* is positioned on the other surface opposite to the second outer surface **112***a* of the second holder **112**. In addition, the second inner surface **112***b* is arranged to face the outer circumferential surface of the arc chamber **21** with the third magnet **123** and the fourth magnet **124** interposed therebetween. In an embodiment, the second inner surface **112***b* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0259] The second inner surface **112***b* is coupled to the third magnet **123** and the fourth magnet **124** of the magnet unit **120**, which will be described later.
[0260] The magnet unit **120** forms a magnetic field inside the arc chamber **21** in which the fixed contact **22** and the movable contact **43** are accommodated. In addition, the fixed contact **22** and the movable contact **43** are located radially inside the magnet unit **120**. In the illustrated embodiment, the magnet unit **120** is arranged so that its center corresponds to the central portion C of the fixed contact **22** and the movable contact **43**.
[0261] The magnet unit **120** may form a magnetic field by itself or with each other. The magnetic field formed by the magnet unit **120** forms electromagnetic force together with the electric current energizing through the fixed contact **22** and the movable contact **43**. The formed electromagnetic force guides an arc generated when the fixed contact **22** and the movable contact **43** are spaced apart.
[0262] In this case, the arc path formation unit **100** forms electromagnetic force in a direction away from the central portion C of the fixed contact **22** and the movable contact **43**. Accordingly, the arc path A.P is also formed in a direction away from the central portion C of the fixed contact **22** and the movable contact **43**.
[0263] As a result, each component provided in the direct current relay **1** may not be damaged by the generated arc. Furthermore, the generated arc can be quickly discharged to the outside of the arc chamber **21**.

[0264] The magnet unit **120** is coupled to the inner surfaces **111***b* and **112***b* of the magnet holder unit **110**. In an embodiment, a fastening member (not shown) may be provided to couple the magnet unit **120** and the inner surfaces **111***b* and **112***b* of the magnet holder unit **110**.
[0265] The magnet unit **120** may include a plurality of magnets.
[0266] In the present embodiment, the magnet unit **120** includes a total of four magnets, including a first magnet **121**, a second magnet **122**, a third magnet **123**, and a fourth magnet **124**.
[0267] The first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** may each be provided in any shape capable of being magnetized to form a magnetic field inside the arc chamber **21**. In addition, the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** are all formed to have polarity in the longitudinal direction and width direction.
[0268] The first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** are arranged to be spaced apart from each other. That is, an empty space is formed between the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124**. In addition, the space between the first magnet **121** and the fourth magnet **124** or the space between the second magnet **122** and the third magnet **123** may function as a passage through which the arc generated in the arc chamber **21** is discharged.
[0269] The first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** may be in contact with or fixedly coupled to the outer circumferential surface of the arc chamber **21**. Accordingly, the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** are preferably formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0270] In an embodiment, the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** may be formed in shapes that correspond to each other. Specifically, the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** may be formed in shapes whose lengths in the width direction and breadth direction respectively correspond to each other.
[0271] The first magnet **121** is coupled to the first inner surface **111***b* of the first holder **111**. In addition, the first magnet **121** extends from one end of the first holder **111** along the first inner surface **111***b*. In an embodiment, the first magnet **121** is formed in a shape corresponding to the first inner surface **111***b* of the first holder **111**.
[0272] The first magnet **121** includes a first facing surface **121***a* and a first opposite surface **121***b*.
[0273] The first facing surface **121***a* is located on one surface of the first magnet **121** facing the central portion C of the fixed contact **22** and the movable contact **43**. In addition, the first facing surface **121***a* is disposed adjacent to the outer circumferential surface of the arc chamber **21**. In an embodiment, the first facing surface **121***a* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0274] The first opposite surface **121***b* is positioned on the other surface opposite to the first facing surface **121***a* of the first magnet **121**. In addition, the first opposite surface **121***b* is disposed to face the inner circumferential surface of the upper frame **11** with the first holder **111** interposed therebetween. In an embodiment, the first opposite surface **121***b* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0275] The second magnet **122** is coupled to the first inner surface **111***b* of the first holder **111**. In addition, the second magnet **122** extends along the first inner surface **111***b* from the other end of the first holder **111** opposite to the first magnet **121**. In an embodiment, the second magnet **122** is formed in a shape corresponding to the first inner surface **111***b* of the first holder **111**.
[0276] The extension direction of the second magnet **122** intersects the extension direction of the first magnet **121**. This results from the fact that the first holder **111** coupled with the first magnet **121** and the second magnet **122** is bent and extended at a predetermined angle.
[0277] The second magnet **122** is arranged to be offset from the first magnet **121** without facing

each other, with a virtual line extending along the arrangement direction of the plurality of fixed contacts **22** interposed therebetween.
[0278] The second magnet **122** includes a second facing surface **122***a* and a second opposite surface **122***b*.
[0279] The second facing surface **122***a* is located on one surface of the second magnet **122** facing the central portion C of the fixed contact **22** and the movable contact **43**. In addition, the second facing surface **122***a* is disposed adjacent to the outer circumferential surface of the arc chamber **21**. In an embodiment, the second facing surface **122***a* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0280] The second opposite surface **122***b* is positioned on the other surface opposite to the second facing surface **122***a* of the second magnet **122**. In addition, the second opposite surface **122***b* is disposed to face the inner circumferential surface of the upper frame **11** with the first holder **111** interposed therebetween. In an embodiment, the second opposite surface **122***b* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0281] The third magnet **123** is coupled to the second inner surface **112***b* of the second holder **112**. In addition, the third magnet **123** extends along the second inner surface **112***b* from one end of the second holder **112** facing the second magnet **122**. In an embodiment, the third magnet **123** is formed in a shape corresponding to the second inner surface **112***b* of the second holder **112**. In the illustrated embodiment, the third magnet **123** extends in a direction parallel to the extension direction of the first magnet **121**.
[0282] The third magnet **123** is arranged to be offset from the first magnet **121** without facing each other with respect to the central portion C of the fixed contact **22** and the movable contact **43**.
[0283] The third magnet **123** includes a third facing surface **123***a* and a third opposite surface **123***b*.
[0284] The third facing surface **123***a* is located on one surface of the third magnet **123** facing the central portion C of the fixed contact **22** and the movable contact **43**. In addition, the third facing surface **123***a* is disposed adjacent to the outer circumferential surface of the arc chamber **21**. In an embodiment, the third facing surface **123***a* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0285] The third opposite surface **123***b* is positioned on the other surface opposite to the third facing surface **123***a* of the third magnet **123**. In addition, the third opposite surface **123***b* is disposed to face the inner circumferential surface of the upper frame **11** with the second holder **112** interposed therebetween. In an embodiment, the third opposite surface **123***b* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0286] The fourth magnet **124** is coupled to the second inner surface **112***b* of the second holder **112**. In addition, the fourth magnet **124** extends along the second inner surface **112***b* from the other end of the second holder **112** facing the first magnet **121** opposite to the third magnet **123**. In an embodiment, the fourth magnet **124** is formed in a shape corresponding to the second inner surface **112***b* of the second holder **112**. In the illustrated embodiment, the fourth magnet **124** extends in a direction parallel to the extension direction of the second magnet **122**.
[0287] The extension direction of the fourth magnet **124** intersects the extension direction of the third magnet **123**. This results from the fact that the second holder **112** coupled with the third magnet **123** and the fourth magnet **124** is bent and extended at a predetermined angle.
[0288] The fourth magnet **124** is arranged to be offset from the third magnet **123** without facing each other, with a virtual line extending along the arrangement direction of the plurality of fixed contacts **22** interposed therebetween.
[0289] The fourth magnet **124** is arranged to be offset from the second magnet **122** without facing each other with respect to the central portion C of the fixed contact **22** and the movable contact **43**.
[0290] In an embodiment, the shortest distance between the third magnet **123** and the fourth magnet **124** is formed to be the same as the shortest distance between the first magnet **121** and the second magnet **122**.

[0291] The fourth magnet **124** includes a fourth facing surface **124***a* and a fourth opposite surface **124***b*.
[0292] The fourth facing surface **124***a* is located on one surface of the fourth magnet **124** facing the central portion C of the fixed contact **22** and the movable contact **43**. In addition, the fourth facing surface **124***a* is disposed adjacent to the outer circumferential surface of the arc chamber **21**. In an embodiment, the fourth facing surface **124***a* is formed in a shape corresponding to the outer circumferential surface of the arc chamber **21**.
[0293] The fourth opposite surface **124***b* is positioned on the other surface opposite to the fourth facing surface **124***a* of the fourth magnet **124**. In addition, the fourth opposite surface **124***b* is disposed to face the inner circumferential surface of the upper frame **11** with the second holder **112** interposed therebetween. In an embodiment, the fourth opposite surface **124***b* is formed in a shape corresponding to the inner circumferential surface of the upper frame **11**.
[0294] In an embodiment, each of the facing surfaces **121***a*, **122***a*, **123***a*, and **124***a* of the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** is all magnetized to the same polarity. Each of the opposite surfaces **121***b*, **122***b*, **123***b*, and **124***b* of the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** is magnetized to polarities opposite to those of each of the facing surfaces **121***a*, **122***a*, **123***a*, and **124***a*, and thus, likewise, all are magnetized to the same polarity.
[0295] In another embodiment, each facing surface **121***a*, **122***a* of the first magnet **121** and the second magnet **122** is magnetized to any one polarity of the N pole and the S pole, and each facing surface **123***a*, **124***a* of the third magnet **123** and the fourth magnet **124** is magnetized to the other one polarity of the N pole and the S pole.
[0296] In addition, the shortest distance from each of the facing surfaces **121***a*, **122***a*, **123***a*, and **124***a* of the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** to the central portion C of the fixed contact **22** and the movable contact **43** may all be formed the same.
[0297] In addition, the shortest path between the first magnet **121** and the third magnet **123** and the shortest path between the second magnet **122** and the fourth magnet **124** overlap the central portion C of the fixed contact **22** and the movable contact **43** in a movement direction of the movable contact **43**.
[0298] Referring to FIGS. **3** to **5**, each of the facing surfaces **121***a*, **122***a*, **123***a*, and **124***a* of the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** is all magnetized to the N pole, and each of the opposite surfaces **121***b*, **122***b*, **123***b*, and **124***b* is all magnetized to the S pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124**.
[0299] In addition, the first holder **111** and the second holder **112** are also magnetized together by the magnet unit **120** to form an additional magnetic field.
[0300] In the embodiment illustrated in FIG. **4**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0301] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the left.
[0302] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.

[0303] In the embodiment illustrated in FIG. **5**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0304] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left.
[0305] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the left.
[0306] Referring to FIGS. **6** to **8**, each of the facing surfaces **121***a*, **122***a*, **123***a*, and **124***a* of the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124** is all magnetized to the S pole, and each of the opposite surfaces **121***b*, **122***b*, **123***b*, and **124***b* is all magnetized to the N pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **121**, the second magnet **122**, the third magnet **123**, and the fourth magnet **124**.
[0307] In addition, the first holder **111** and the second holder **112** are also magnetized together by the magnet unit **120** to form an additional magnetic field.
[0308] In the embodiment illustrated in FIG. **7**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0309] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward and to the left.
[0310] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the left.
[0311] In the embodiment illustrated in FIG. **8**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0312] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left.
[0313] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0314] Referring to FIGS. **9** to **11**, each of the facing surfaces **121***a* and **122***a* of the first magnet **121** and the second magnet **122** is all magnetized to the N pole, and each of the facing surfaces **123***a* and **124***a* of the third magnet **123** and the fourth magnet **124** is all magnetized to the S pole.
[0315] Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **121** and the second magnet **122** and between the third magnet **123** and the fourth magnet **124**. On the contrary, a magnetic field in a direction from the first magnet **121** toward the third magnet **123** and the fourth magnet **124** is formed between the first magnet **121**, the third

magnet **123**, and the fourth magnet **124**. In addition, a magnetic field in a direction from the second magnet **122** toward the third magnet **123** and the fourth magnet **124** is formed between the second magnet **122**, the third magnet **123**, and the fourth magnet **124**.

[0316] In addition, the first holder **111** and the second holder **112** are also magnetized together by the magnet unit **120** to form an additional magnetic field.

[0317] In the embodiment illustrated in FIG. **10**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.

[0318] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the left.

[0319] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.

[0320] In the embodiment illustrated in FIG. **11**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.

[0321] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face to the left.

[0322] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face to the right.

[0323] Therefore, the arc path formation unit **100** according to the present embodiment may form the electromagnetic force and the arc path A.P in a direction away from the central portion C, regardless of the polarity of the magnet unit **120** or the direction of the electric current energizing through the direct current relay.

[0324] Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central portion C can be prevented. Furthermore, the generated arc can be quickly discharged to the outside, so that the operation reliability of the direct current relay **1** can be improved.

3. Description of the Arc Path Formation Unit **200** According to the Second Embodiment of the Present Disclosure

[0325] Hereinafter, the arc path formation unit **200** according to the second embodiment of the present disclosure will be described with reference to FIGS. **12** to **19**.

[0326] The arc path formation unit **200** according to the present embodiment includes a magnet holder unit **210** and a magnet unit **220**.

[0327] The magnet holder unit **210** according to the present embodiment has the same structure and function as the magnet holder unit **110** according to the above-described embodiment. However, the magnet unit **220** according to the present embodiment differs from the magnet unit **120** according to the above-described embodiment in that the first magnet **221** and the third magnet **223** are arranged to face the second magnet **222** and the fourth magnet **224** each other, respectively, with a virtual line extending along the arrangement direction of the fixed contact **22** interposed therebetween.

[0328] Thus, the description of the magnet holder unit **210** will be replaced by the description of the magnet holder unit **110** according to the above-described embodiment, and the magnet unit **220**

will be described focusing on a difference from the magnet unit **120** according to the above-described embodiment.
[0329] The magnet unit **220** according to the present embodiment includes a first magnet **221**, a second magnet **222**, a third magnet **223**, and a fourth magnet **224**.
[0330] The first magnet **221** is arranged to face the second magnet **222** each other, with a virtual line extending along the arrangement direction of the fixed contact **22** interposed therebetween.
[0331] The third magnet **223** is arranged to face the fourth magnet **224** each other, with a virtual line extending along the arrangement direction of the fixed contact **22** interposed therebetween.
[0332] In an embodiment, the shortest distance from each of the facing surfaces **221***a*, **222***a*, **223***a*, and **224***a* of the first magnet **221**, the second magnet **222**, the third magnet **223**, and the fourth magnet **224** to the center of the auxiliary magnet **230** may all be formed the same.
[0333] Referring to FIGS. **12** to **14**, each of the facing surfaces **221***a*, **222***a*, **223***a*, and **224***a* of the first magnet **221**, the second magnet **222**, the third magnet **223**, and the fourth magnet **224** is all magnetized to the N pole, and each of the opposite surfaces **221***b*, **222***b*, **223***b*, and **224***b* is all magnetized to the S pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **221**, the second magnet **222**, the third magnet **223**, and the fourth magnet **224**.
[0334] In addition, the first holder **211** and the second holder **212** are also magnetized together by the magnet unit **220** to form an additional magnetic field.
[0335] In the embodiment illustrated in FIG. **13**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0336] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the left.
[0337] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0338] In the embodiment illustrated in FIG. **14**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0339] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left.
[0340] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0341] Referring to FIGS. **15** to **17**, each of the facing surfaces **221***a*, **222***a*, **223***a*, and **224***a* of the first magnet **221**, the second magnet **222**, the third magnet **223**, and the fourth magnet **224** is all magnetized to the S pole, and each of the opposite surfaces **221***b*, **222***b*, **223***b*, and **224***b* is all magnetized to the N pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **221**, the second magnet **222**, the third magnet **223**, and the fourth magnet **224**.
[0342] In addition, the first holder **211** and the second holder **212** are also magnetized together by

the magnet unit **220** to form an additional magnetic field.
[0343] In the embodiment illustrated in FIG. **16**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0344] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward and to the left.
[0345] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0346] In the embodiment illustrated in FIG. **17**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0347] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left.
[0348] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0349] Referring to FIGS. **18** to **19**, each of the facing surfaces **221***a* and **222***a* of the first magnet **221** and the second magnet **222** is all magnetized to the N pole, and each of the facing surfaces **223***a* and **224***a* of the third magnet **223** and the fourth magnet **224** is all magnetized to the S pole.
[0350] Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **221** and the second magnet **222** and between the third magnet **223** and the fourth magnet **224**. On the contrary, a magnetic field in a direction from the first magnet **221** toward the third magnet **223** and the fourth magnet **224** is formed between the first magnet **221**, the third magnet **223**, and the fourth magnet **224**. In addition, a magnetic field in a direction from the second magnet **222** toward the third magnet **223** and the fourth magnet **224** is formed between the second magnet **222**, the third magnet **223**, and the fourth magnet **224**.
[0351] In addition, the first holder **211** and the second holder **212** are also magnetized together by the magnet unit **220** to form an additional magnetic field.
[0352] In the embodiment illustrated in FIG. **19**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a* or a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0353] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward and to the left.
[0354] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.

[0355] Therefore, the arc path formation unit **200** according to the present embodiment may form the electromagnetic force and the arc path A.P in a direction away from the central portion C, regardless of the polarity of the magnet unit **220** or the direction of the electric current energizing through the direct current relay.
[0356] Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central portion C can be prevented. Furthermore, the generated arc can be quickly discharged to the outside, so that the operation reliability of the direct current relay **1** can be improved.
4. Description of the Arc Path Formation Unit **300** According to the Third Embodiment of the Present Disclosure
[0357] Hereinafter, the arc path formation unit **300** according to the third embodiment of the present disclosure will be described with reference to FIGS. **20** to **31**.
[0358] The arc path formation unit **300** according to the present embodiment includes a magnet holder unit **310** and a magnet unit **320**.
[0359] The magnet holder unit **310** according to the present embodiment has the same structure and function as the magnet holder unit **210** according to the above-described embodiment. However, the magnet unit **320** according to the present embodiment differs from the magnet unit **220** according to the above-described embodiment in that at least two of the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324** are formed in different sizes.
[0360] Thus, the description of the magnet holder unit **310** will be replaced by the description of the magnet holder unit **210** according to the above-described embodiment, and the magnet unit **320** will be described focusing on a difference from the magnet unit **220** according to the above-described embodiment.
[0361] The magnet unit **320** according to the present embodiment includes a first magnet **321**, a second magnet **322**, a third magnet **323**, and a fourth magnet **324**.
[0362] At least two of the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324** are formed in different sizes.
[0363] In an embodiment, the first magnet **321** may be formed to have a first length in the longitudinal direction, and the second magnet **322**, the third magnet **323**, and the fourth magnet **324** may be formed to have a second length in the longitudinal direction.
[0364] In the illustrated embodiment, the first magnet **321** and the third magnet **323** are formed in different sizes, and the second magnet **322** and the fourth magnet **324** are also formed in different sizes.
[0365] In an embodiment, the first magnet **321** and the second magnet **322** may be formed to have a first length in the longitudinal direction, and the third magnet **323** and the fourth magnet **324** may be formed to have a second length in the longitudinal direction.
[0366] The second magnet **322** is arranged to face the first magnet **321** each other, with a virtual line extending along the arrangement direction of the fixed contact **22** interposed therebetween.
[0367] The fourth magnet **324** is symmetrical to the second magnet **322** with respect to the central portion C of the fixed contact **22** and the movable contact **43**.
[0368] In addition, the fourth magnet **324** is arranged to face the third magnet **323** each other, with a virtual line extending along the arrangement direction of the fixed contact **22** interposed therebetween.
[0369] Referring to FIGS. **20** to **22**, each of the facing surfaces **321***a*, **322***a*, **323***a*, and **324***a* of the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324** is all magnetized to the N pole, and each of the opposite surfaces **321***b*, **322***b*, **323***b*, and **324***b* is all magnetized to the S pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324**.
[0370] In addition, the first holder **311** and the second holder **312** are also magnetized together by the magnet unit **320** to form an additional magnetic field.

[0371] In the embodiment illustrated in FIG. **21**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0372] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the left.
[0373] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0374] In the embodiment illustrated in FIG. **22**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0375] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left.
[0376] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0377] Referring to FIGS. **23** to **25**, each of the facing surfaces **321***a*, **322***a*, **323***a*, and **324***a* of the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324** is all magnetized to the S pole, and each of the opposite surfaces **321***b*, **322***b*, **323***b*, and **324***b* is all magnetized to the N pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **321**, the second magnet **322**, the third magnet **323**, and the fourth magnet **324**.
[0378] In addition, the first holder **311** and the second holder **312** are also magnetized together by the magnet unit **320** to form an additional magnetic field.
[0379] In the embodiment illustrated in FIG. **24**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0380] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward and to the left.
[0381] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0382] In the embodiment illustrated in FIG. **25**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0383] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left.

[0384] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0385] Referring to FIGS. **26** to **28**, each of the facing surfaces **321***a* and **322***a* of the first magnet **321** and the second magnet **322** is all magnetized to the N pole, and each of the facing surfaces **323***a* and **324***a* of the third magnet **323** and the fourth magnet **324** is all magnetized to the S pole.
[0386] Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **321** and the second magnet **322** and between the third magnet **323** and the fourth magnet **324**. On the contrary, a magnetic field in a direction from the first magnet **321** toward the third magnet **323** and the fourth magnet **324** is formed between the first magnet **321**, the third magnet **323**, and the fourth magnet **324**. In addition, a magnetic field in a direction from the second magnet **322** toward the third magnet **323** and the fourth magnet **324** is formed between the second magnet **322**, the third magnet **323**, and the fourth magnet **324**.
[0387] In addition, the first holder **311** and the second holder **312** are also magnetized together by the magnet unit **320** to form an additional magnetic field.
[0388] In the embodiment illustrated in FIG. **27**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0389] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward.
[0390] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0391] In the embodiment illustrated in FIG. **28**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0392] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward.
[0393] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0394] Referring to FIGS. **29** to **31**, each of the facing surfaces **321***a* and **322***a* of the first magnet **321** and the second magnet **322** is all magnetized to the S pole, and each of the facing surfaces **323***a* and **324***a* of the third magnet **323** and the fourth magnet **324** is all magnetized to the N pole.
[0395] Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **321** and the second magnet **322** and between the third magnet **323** and the fourth magnet **324**. On the contrary, a magnetic field in a direction from the third magnet **323** and the fourth magnet **324** toward the first magnet **321** is formed between the first magnet **321**, the third magnet **323**, and the fourth magnet **324**. In addition, a magnetic field in a direction from the third magnet **323** and the fourth magnet **324** toward the second magnet **322** is formed between the second magnet **322**, the third magnet **323**, and the fourth magnet **324**.
[0396] In addition, the first holder **311** and the second holder **312** are also magnetized together by

the magnet unit **320** to form an additional magnetic field.
[0397] In the embodiment illustrated in FIG. **30**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0398] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward.
[0399] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the left.
[0400] In the embodiment illustrated in FIG. **31**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0401] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face upward.
[0402] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the left.
[0403] Therefore, the arc path formation unit **300** according to the present embodiment may form the electromagnetic force and the arc path A.P in a direction away from the central portion C, regardless of the polarity of the magnet unit **320** or the direction of the electric current energizing through the direct current relay.
[0404] Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central portion C can be prevented. Furthermore, the generated arc can be quickly discharged to the outside, so that the operation reliability of the direct current relay **1** can be improved.
5. Description of the Arc Path Formation Unit **400** According to the Fourth Embodiment of the Present Disclosure
[0405] Hereinafter, the arc path formation unit **400** according to the fourth embodiment of the present disclosure will be described with reference to FIGS. **32** to **40**.
[0406] The arc path formation unit **400** according to the present embodiment includes a magnet holder unit **410** and a magnet unit **420**.
[0407] The magnet holder unit **410** according to the present embodiment has the same structure and function as the magnet holder unit **310** according to the above-described embodiment. However, the magnet unit **420** according to the present embodiment differs from the magnet unit **320** according to the above-described embodiment in that the first magnet **421** and the second magnet **422** are formed in different sizes, and the third magnet **423** and the fourth magnet **424** are formed in different sizes.
[0408] Thus, the description of the magnet holder unit **410** will be replaced by the description of the magnet holder unit **310** according to the above-described embodiment, and the magnet unit **420** will be described focusing on a difference from the magnet unit **320** according to the above-described embodiment.
[0409] The magnet unit **420** according to the present embodiment includes a first magnet **421**, a second magnet **422**, a third magnet **423**, and a fourth magnet **424**.
[0410] The first magnet **421** and the second magnet **422** are formed in different sizes. In addition, the third magnet **423** and fourth magnet **424** are formed in different sizes.

[0411] In an embodiment, the first magnet **421** and the third magnet **423** are formed to have a first length in the longitudinal direction, and the second magnet **422** and the fourth magnet **424** are formed to have a second length in the longitudinal direction.
[0412] The third magnet **423** is symmetrical to the first magnet **421** with respect to the central portion C of the fixed contact **22** and the movable contact **43**.
[0413] The fourth magnet **424** is symmetrical to the second magnet **422** with respect to the central portion C of the fixed contact **22** and the movable contact **43**.
[0414] Referring to FIGS. **32** to **34**, each of the facing surfaces **421***a*, **422***a*, **423***a*, and **424***a* of the first magnet **421**, the second magnet **422**, the third magnet **423**, and the fourth magnet **424** is all magnetized to the N pole, and each of the opposite surfaces **421***b*, **422***b*, **423***b*, and **424***b* is all magnetized to the S pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **421**, the second magnet **422**, the third magnet **423**, and the fourth magnet **424**.
[0415] In addition, the first holder **411** and the second holder **412** are also magnetized together by the magnet unit **420** to form an additional magnetic field.
[0416] In the embodiment illustrated in FIG. **33**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0417] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face downward and to the left.
[0418] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0419] In the embodiment illustrated in FIG. **34**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0420] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face upward and to the left.
[0421] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0422] Referring to FIGS. **35** to **37**, each of the facing surfaces **421***a*, **422***a*, **423***a*, and **424***a* of the first magnet **421**, the second magnet **422**, the third magnet **423**, and the fourth magnet **424** is all magnetized to the S pole, and each of the opposite surfaces **421***b*, **422***b*, **423***b*, and **424***b* is all magnetized to the N pole. Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **421**, the second magnet **422**, the third magnet **423**, and the fourth magnet **424**.
[0423] In addition, the first holder **411** and the second holder **412** are also magnetized together by the magnet unit **420** to form an additional magnetic field.
[0424] In the embodiment illustrated in FIG. **36**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0425] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in

the vicinity of the first fixed contact **22***a* is formed to face upward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the left.
[0426] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0427] In the embodiment illustrated in FIG. **37**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0428] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left.
[0429] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the right.
[0430] Referring to FIGS. **38** to **40**, each of the facing surfaces **421***a* and **422***a* of the first magnet **421** and the second magnet **422** is all magnetized to the N pole, and each of the facing surfaces **423***a* and **424***a* of the third magnet **423** and the fourth magnet **424** is all magnetized to the S pole.
[0431] Accordingly, a magnetic field in a direction of pushing each other is formed between the first magnet **421** and the second magnet **422** and between the third magnet **423** and the fourth magnet **424**. On the contrary, a magnetic field in a direction from the first magnet **421** toward the third magnet **423** and the fourth magnet **424** is formed between the first magnet **421**, the third magnet **423**, and the fourth magnet **424**. In addition, a magnetic field in a direction from the second magnet **422** toward the third magnet **423** and the fourth magnet **424** is formed between the second magnet **422**, the third magnet **423**, and the fourth magnet **424**.
[0432] In addition, the first holder **411** and the second holder **412** are also magnetized together by the magnet unit **420** to form an additional magnetic field.
[0433] In the embodiment illustrated in FIG. **39**, the direction of the electric current is a direction from the second fixed contact **22***b* through the movable contact **43** to the first fixed contact **22***a*.
[0434] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is also formed to face upward and to the right.
[0435] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face upward and to the right. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face upward and to the right.
[0436] In the embodiment illustrated in FIG. **40**, the direction of the electric current is a direction from the first fixed contact **22***a* through the movable contact **43** to the second fixed contact **22***b*.
[0437] If Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the first fixed contact **22***a*, the electromagnetic force generated in the vicinity of the first fixed contact **22***a* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the first fixed contact **22***a* is formed to face downward and to the

left.

[0438] Likewise, if Fleming's left-hand rule is applied considering the direction of the electric current and the direction of the magnetic field in the second fixed contact **22***b*, the electromagnetic force generated in the vicinity of the second fixed contact **22***b* is formed to face downward and to the left. Accordingly, the arc path A.P in the vicinity of the second fixed contact **22***b* is also formed to face downward and to the left.

[0439] Therefore, the arc path formation unit **400** according to the present embodiment may form the electromagnetic force and the arc path A.P in a direction away from the central portion C, regardless of the polarity of the magnet unit **420** or the direction of the electric current energizing through the direct current relay.

[0440] Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central portion C can be prevented. Furthermore, the generated arc can be quickly discharged to the outside, so that the operation reliability of the direct current relay **1** can be improved.

[0441] Although the present disclosure has been described above with reference to preferred exemplary embodiments thereof, the present disclosure is not limited to the configurations of the above-described embodiments.

[0442] In addition, the present disclosure may be variously modified and changed without departing from the idea and scope of the present disclosure described in the following claims by those skilled in the art to which the present disclosure pertains.

[0443] Furthermore, the embodiments may be configured by selectively combining all or some of the embodiments so that various modifications may be made thereto.

DESCRIPTION OF SYMBOLS

[0444] **1**: direct current relay [0445] **10**: frame unit [0446] **11**: upper frame [0447] **12**: lower frame [0448] **13**: insulating plate [0449] **14**: support plate [0450] **20**: switch unit [0451] **21**: arc chamber [0452] **22**: fixed contact [0453] **22***a:* first fixed contact [0454] **22***b:* second fixed contact [0455] **30**: core unit [0456] **31**: stationary core [0457] **32**: movable core [0458] **33**: yoke [0459] **34**: bobbin [0460] **35**: coil [0461] **36**: return spring [0462] **37**: cylinder [0463] **40**: movable contact unit [0464] **41**: housing [0465] **42**: cover [0466] **43**: movable contact [0467] **44**: shaft [0468] **45**: clastic portion [0469] **100**: first embodiment of arc path formation unit [0470] **110**: magnet holder unit [0471] **111**: first holder [0472] **111***a*: first outer surface [0473] **111***b*: first inner surface [0474] **112**: second holder [0475] **112***a*: second outer surface [0476] **112***b*: second inner surface [0477] **120**: magnet unit [0478] **121**: first magnet [0479] **121***a*: first facing surface [0480] **121***b*: first opposite surface [0481] **122**: second magnet [0482] **122***a*: second facing surface [0483] **122***b*: second opposite surface [0484] **123**: third magnet [0485] **123***a*: third facing surface [0486] **123***b*: third opposite surface [0487] **124**: fourth magnet [0488] **124***a*: fourth facing surface [0489] **124***b*: fourth opposite surface [0490] **200**: second embodiment of arc path formation unit [0491] **210**: magnet holder unit [0492] **211**: first holder [0493] **211***a*: first outer surface [0494] **211***b*: first inner surface [0495] **212**: second holder [0496] **212***a*: second outer surface [0497] **212***b*: second inner surface [0498] **220**: magnet unit [0499] **221**: first magnet [0500] **221***a*: first facing surface [0501] **221***b*: first opposite surface [0502] **222**: second magnet [0503] **222***a*: second facing surface [0504] **222***b*: second opposite surface [0505] **223**: third magnet [0506] **223***a*: third facing surface [0507] **223***b*: third opposite surface [0508] **224**: fourth magnet [0509] **224***a*: fourth facing surface [0510] **224***b*: fourth opposite surface [0511] **300**: third embodiment of arc path formation unit [0512] **310**: magnet holder unit [0513] **311**: first holder [0514] **311***a*: first outer surface [0515] **311***b*: first inner surface [0516] **312**: second holder [0517] **312***a*: second outer surface [0518] **312***b*: second inner surface [0519] **320**: magnet unit [0520] **321**: first magnet [0521] **321***a*: first facing surface [0522] **321***b*: first opposite surface [0523] **322**: second magnet [0524] **322***a*: second facing surface [0525] **322***b*: second opposite surface [0526] **323**: third magnet [0527] **323***a*: third facing surface [0528] **323***b*: third opposite surface [0529] **324**: fourth magnet [0530] **324***a*: fourth facing surface [0531] **324***b*: fourth opposite surface [0532] **400**: fourth embodiment of arc path formation unit [0533] **410**:

magnet holder unit [0534] **411**: first holder [0535] **411***a*: first outer surface [0536] **411***b*: first inner surface [0537] **412**: second holder [0538] **412***a*: second outer surface [0539] **412***b*: second inner surface [0540] **420**: magnet unit [0541] **421**: first magnet [0542] **421***a*: first facing surface [0543] **421***b*: first opposite surface [0544] **422**: second magnet [0545] **422***a*: second facing surface [0546] **422***b*: second opposite surface [0547] **423**: third magnet [0548] **423***a*: third facing surface [0549] **423***b*: third opposite surface [0550] **424**: fourth magnet [0551] **424***a*: fourth facing surface [0552] **424***b*: fourth opposite surface [0553] A.P: arc path

Claims

1. An arc path formation unit, comprising: an arc chamber in which a plurality of fixed contacts and movable contacts are accommodated; a magnet holder unit disposed outside the arc chamber and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit comprises: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein the first magnet and the second magnet are arranged to be offset from the third magnet and the fourth magnet, respectively, without facing each other, with respect to the central point of the plurality of fixed contacts.
2. The arc path formation unit of claim 1, wherein a shortest path between the first magnet and the third magnet overlaps the central point of the plurality of fixed contacts in a movement direction of the movable contact, and a shortest path between the second magnet and the fourth magnet overlaps the central point of the plurality of fixed contacts in a movement direction of the movable contact.
3. The arc path formation unit of claim 1, wherein the first magnet extends in a direction parallel to an extension direction of the third magnet, and the second magnet extends in a direction parallel to an extension direction of the fourth magnet.
4. The arc path formation unit of claim 3, wherein the first magnet and the second magnet have their respective extension directions intersecting each other.
5. The arc path formation unit of claim 1, wherein the first magnet is arranged to be offset from the second magnet without facing each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet is arranged to be offset from the fourth magnet without facing each other, with the virtual line interposed therebetween.
6. The arc path formation unit of claim 5, wherein in the magnet unit, a shortest distance between the first magnet and the second magnet is formed to be equal to a shortest distance between the third magnet and the fourth magnet.
7. The arc path formation unit of claim 1, wherein the first magnet is arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet is arranged to face the fourth magnet each other with the virtual line interposed therebetween.
8. The arc path formation unit of claim 7, wherein in the magnet unit, a shortest distance between the first magnet and the second magnet is formed to be equal to a shortest distance between the third magnet and the fourth magnet.

9. The arc path formation unit of claim 1, wherein the first magnet, the second magnet, the third magnet, and the fourth magnet are all magnetized to the same polarity.
10. The arc path formation unit of claim 1, wherein the first magnet and the second magnet are magnetized to any one polarity of the N pole and the S pole, and the third magnet and the fourth magnet are magnetized to the other one polarity of the N pole and the S pole.
11. An arc path formation unit, comprising: an arc chamber in which a plurality of fixed contacts and movable contacts are accommodated; a magnet holder unit disposed outside the arc chamber and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit comprises: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein at least two of the first magnet, the second magnet, the third magnet, and the fourth magnet are formed in different sizes.
12. The arc path formation unit of claim 11, wherein the first magnet and the third magnet are formed in different sizes, and the second magnet and the fourth magnet are also formed in different sizes.
13. The arc path formation unit of claim 12, wherein the first magnet and the second magnet are formed to have a first length in the longitudinal direction, and the third magnet and the fourth magnet are formed to have a second length in the longitudinal direction.
14. The arc path formation unit of claim 13, wherein the first magnet is arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet is arranged to face the fourth magnet each other with the virtual line interposed therebetween.
15. The arc path formation unit of claim 11, wherein the first magnet and the second magnet are formed in different sizes, and the third magnet and the fourth magnet are also formed in different sizes.
16. The arc path formation unit of claim 15, wherein the first magnet and the third magnet are formed to have a first length in the longitudinal direction, and the second magnet and the fourth magnet are formed to have a second length in the longitudinal direction.
17. The arc path formation unit of claim 16, wherein the first magnet is symmetrical to the third magnet with respect to a central point of the plurality of fixed contacts, and the second magnet is symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts.
18. The arc path formation unit of claim 11, wherein the first magnet is formed to have a first length in the longitudinal direction, and the second magnet, the third magnet, and the fourth magnet are formed to have a second length in the longitudinal direction.
19. The arc path formation unit of claim 18, wherein the second magnet is symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts.
20. The arc path formation unit of claim 18, wherein the third magnet is arranged to face the fourth magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween.
21. The arc path formation unit of claim 11, wherein the first magnet, the second magnet, the third magnet, and the fourth magnet are all magnetized to the same polarity.
22. The arc path formation unit of claim 11, wherein the first magnet and the second magnet are

magnetized to any one polarity of the N pole and the S pole, and the third magnet and the fourth magnet are magnetized to the other one polarity of the N pole and the S pole.

23. A direct current relay, comprising: a plurality of fixed contacts spaced apart from each other in one direction; a movable contact that is in contact with or spaced apart from the fixed contact; an arc chamber in which a space is formed to accommodate the fixed contact and the movable contact; a frame surrounding the arc chamber; a magnet holder unit disposed between the outside of the arc chamber and the inside of the frame and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit comprises: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein the first magnet and the second magnet are arranged to be offset from the third magnet and the fourth magnet, respectively, without facing each other, with respect to the central point of the plurality of fixed contacts.

24. The direct current relay of claim 23, wherein the first magnet extends in a direction parallel to an extension direction of the third magnet, and the second magnet extends in a direction parallel to an extension direction of the fourth magnet, and its extension direction intersects the extension direction of the first magnet.

25. The direct current relay of claim 23, wherein the first magnet is arranged to be offset from the second magnet without facing each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet is arranged to be offset from the fourth magnet without facing each other, with the virtual line interposed therebetween.

26. The direct current relay of claim 23, wherein the first magnet is arranged to face the second magnet each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and the third magnet is arranged to face the fourth magnet each other with the virtual line interposed therebetween.

27. A direct current relay, comprising: a plurality of fixed contacts spaced apart from each other in one direction; a movable contact that is in contact with or spaced apart from the fixed contact; an arc chamber in which a space is formed to accommodate the fixed contact and the movable contact; a frame surrounding the arc chamber; a magnet holder unit disposed between the outside of the arc chamber and the inside of the frame and comprising a first holder and a second holder that are different from each other; and a magnet unit attached to one surface of the magnet holder unit facing the arc chamber and forming a magnetic field in the arc chamber, wherein the first holder and the second holder are each bent and extended at a predetermined angle, are spaced apart from each other and are arranged in a direction parallel to the arrangement direction of the plurality of fixed contacts, and are arranged such that their respective concave portions face each other, wherein the magnet unit comprises: a first magnet and a second magnet disposed adjacent to one surface of the first holder facing the arc chamber and extending from one end or the other end of the first holder along the one surface of the first holder; and a third magnet and a fourth magnet disposed adjacent to one surface of the second holder facing the arc chamber and extending along the one surface of the second holder from one end of the second holder facing the second magnet or the other end facing the first magnet, and wherein at least two of the first magnet, the second magnet, the third magnet, and the fourth magnet are formed in different sizes.

28. The direct current relay of claim 27, wherein the first magnet and the second magnet are arranged to face each other, with a virtual line extending along the arrangement direction of the fixed contact interposed therebetween, and are formed to have a first length in the longitudinal direction, and wherein the third magnet and the fourth magnet are arranged to face each other, with the virtual line interposed therebetween, and are formed to have a second length in the longitudinal direction.

29. The direct current relay of claim 27, wherein the first magnet is symmetrical to the third magnet with respect to a central point of the plurality of fixed contacts, the second magnet is symmetrical to the fourth magnet with respect to the central point of the plurality of fixed contacts, the first magnet and the third magnet are formed to have a first length in the longitudinal direction, and the second magnet and the fourth magnet are formed to have a second length in the longitudinal direction.

30. The direct current relay of claim 27, wherein the first magnet is formed to have a first length in the longitudinal direction, and the second magnet, the third magnet, and the fourth magnet are formed to have a second length in the longitudinal direction.